Tutorial 7 — Type I & II Errors, Size, and Power

Introductory Econometrics — TA Session

Duration: 50 minutes

Part 1: Concepts *(25 min)*

Block 1: Type I and Type II Errors (10 min)

The Four Possible Outcomes of a Hypothesis Test

Every hypothesis test makes one binary decision—**Reject** H_0 or **Fail to reject** H_0—while the truth is in one of two states. This gives four possible outcomes:

	H_0 **is actually TRUE**	H_0 **is actually FALSE**
Reject H_0	**Type I Error** (false positive)	**Correct!** (Power)
Fail to reject H_0	**Correct!** (true negative)	**Type II Error** (false negative)

Type I Error (α): We reject H_0 when it is true. We see a pattern that is not there.

Type II Error (β): We fail to reject H_0 when it is false. A real effect is present but we miss it.

The asymmetry: In most contexts the two errors are not equally costly. Criminal law ("beyond reasonable doubt") prioritises minimising Type I—convicting an innocent person—even at the cost of more Type II errors. Medical screening may reverse this priority.

Quick Exercise. Fill in the table for the following context:

We test whether a job-training programme has an effect on wages.
*H_0: the programme has **no effect** on wages.*

	Type I Error	Type II Error
What happens		
In words		
Real-world consequence		

Solution

H_0**:** The programme has no effect ($\beta_1 = 0$).

Type I Error: We reject H_0 and conclude the programme works—but it actually does not. We credit a useless programme and waste public resources.

Type II Error: We fail to reject H_0 and conclude the programme is ineffective—but it actually works. A valuable policy is dismissed and workers miss the benefit.

Which is worse? Depends on the stakes. If the programme is cheap to run, Type II is worse (we lose real gains). If funding is scarce, Type I is worse (we fund an ineffective policy).

How to read the figure below

The figure has **two panels**, one for each state of the world.

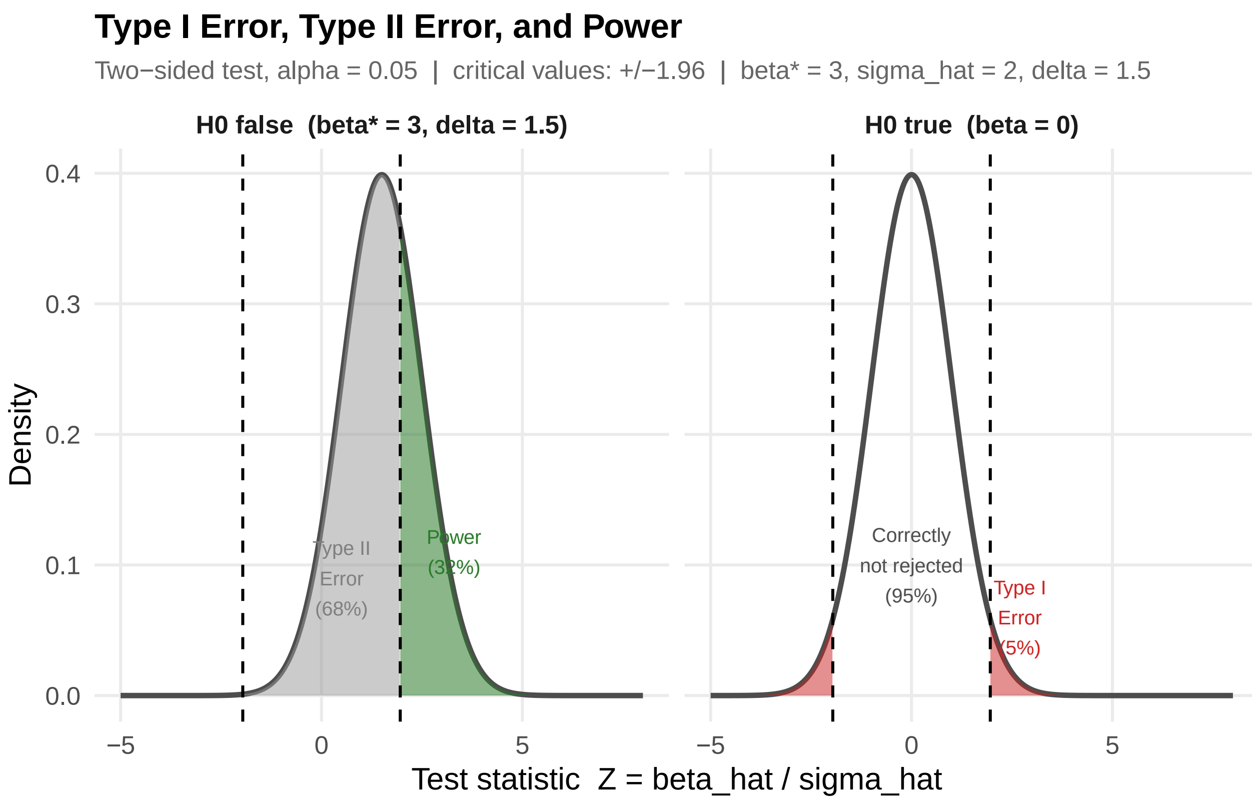
Left panel — H_0 is true (programme has no effect, $\beta_1 = 0$). The bell curve shows all values of the test statistic $Z = \hat{\beta}_1/\sigma_{\hat{\beta}_1}$ we could observe across many repeated studies, when there truly is no effect. The curve is centred at zero. The dashed vertical lines at ± 1.96 are the critical values: we reject H_0 whenever Z falls outside them.

- **Red tails (Type I Error, 5%):** these are the studies where we *would* reject H_0—even though it is true. The programme does nothing, but by bad luck our estimate lands far from zero. This happens 5% of the time. We cannot eliminate it; we chose $\alpha = 0.05$ precisely to accept it.
- **White centre (95%):** the studies where we correctly do not reject H_0.

Right panel — H_0 is false (programme raises wages by $\beta_1^* = 3$). Now the true effect is non-zero, so the distribution of Z shifts to the right—it is no longer centred at zero but at $\delta = \beta_1^*/\sigma_{\hat{\beta}_1} = 1.5$.

- **Green area (Power):** studies where Z exceeds the critical value and we *correctly* reject H_0. The programme works, and we detect it.
- **Grey area (Type II Error):** studies where Z stays inside the critical values and we *fail* to detect the real effect. The distribution has shifted, but not enough for most draws to clear the bar.

Key intuition: the two panels share the same critical values (dashed lines). What changes is the *centre* of the bell curve. When the true effect is small, the right-panel curve barely moves—most of its mass stays inside the critical values, giving low power and high Type II error. A larger true effect shifts the curve further right, putting more mass beyond the threshold.



Block 2: Size (7 min)

Definition of Size

$$\text{Size} = P(\text{reject } H_0 \mid H_0 \text{ is true}) = \alpha$$

The **size** of a test equals the Type I error probability. When we set $\alpha = 0.05$ and use critical value $z_{0.025} = 1.96$, we construct a procedure that—**when** H_0 **is true**—rejects exactly 5% of the time in repeated sampling.

$$P(|Z| > 1.96 \mid H_0) = P(Z > 1.96) + P(Z < -1.96) = 0.025 + 0.025 = 0.05$$

What $\alpha = 0.05$ **means:** if we ran the same experiment 1,000 times in a world where H_0 is true, we would falsely reject H_0 about 50 times—purely by random chance.

What α **does NOT mean:** it is not the probability that H_0 is true. It is a long-run frequency property of the *procedure*, not of any single conclusion.

Block 3: Power (8 min)

Definition of Power

$$\text{Power}(\beta^*) = P(\text{reject } H_0 \mid \beta = \beta^*) = 1 - P(\text{Type II Error} \mid \beta = \beta^*)$$

Power is the probability of **correctly** rejecting H_0 when the true parameter is β^*.

What makes power high?

1. Large true effect $|\beta^*|$: the bigger the signal, the easier to detect.
2. Small $\sigma_{\hat{\beta}}$: more precision means larger signal-to-noise ratio.
3. Larger α: a looser Type I threshold makes rejection easier—at a cost.

Analytical formula. When $Z = \hat{\beta}/\sigma_{\hat{\beta}}$ and the true value is β^*:

$$\text{Power}(\beta^*) = \Phi\left(\frac{|\beta^*|}{\sigma_{\hat{\beta}}} - z_{\alpha/2}\right) + \Phi\left(-\frac{|\beta^*|}{\sigma_{\hat{\beta}}} - z_{\alpha/2}\right) \approx \Phi(\delta - 1.96) \quad \text{for large } \delta$$

where $\delta = |\beta^*|/\sigma_{\hat{\beta}}$ is the **signal-to-noise ratio**.

Key property: Power is a *function* of β^*, not a single number. When $\beta^* = 0$ (i.e., H_0 is exactly true), Power $= \alpha$. As $|\beta^*|$ grows, Power increases toward 1.

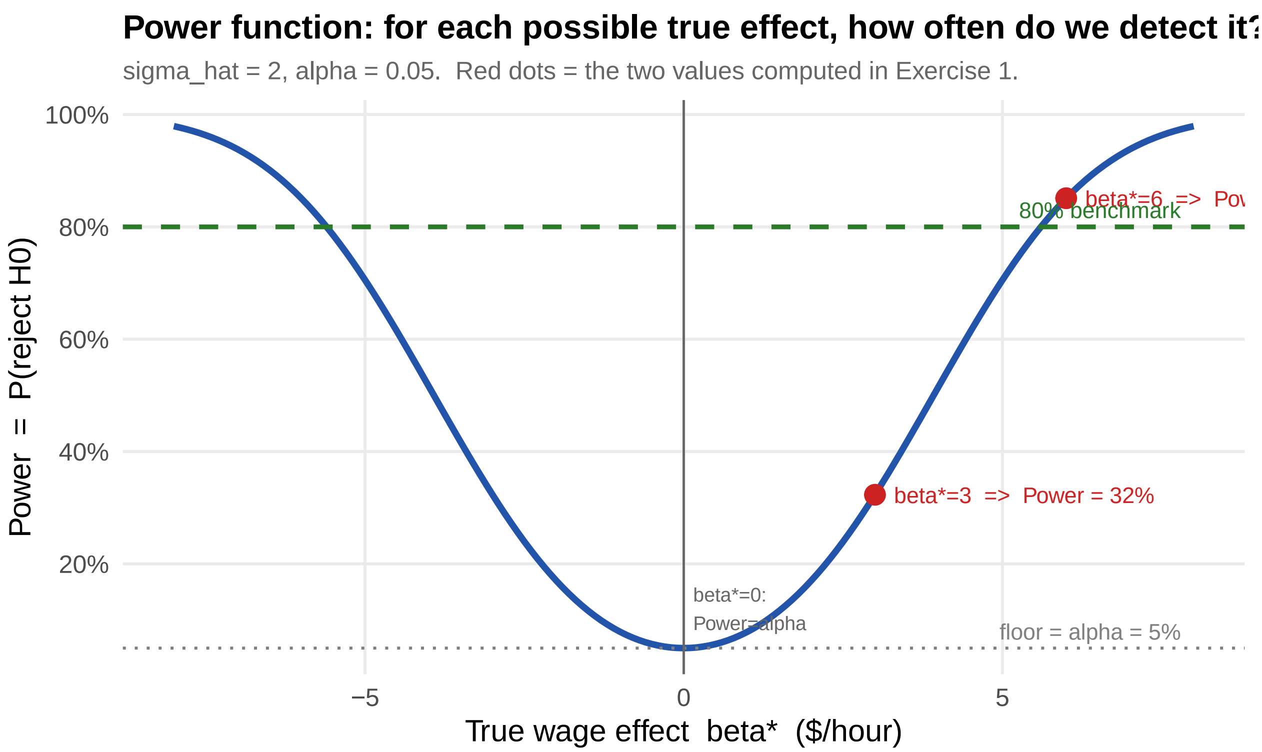
How to read the power curve below

The x-axis is the **true effect** β^*—a value we never observe in practice. The y-axis is the **probability that our test detects it**.

Read the curve like this: "*If the true wage effect of the programme were β^* dollars/hour, how often would our test correctly reject H_0?*"

- **At $\beta^* = 0$:** H_0 is exactly true. Power $= \alpha = 5\%$. We reject only by chance—this is the Type I error floor. The curve never goes below this line.
- **At $\beta^* = 3$:** The programme raises wages by \$3/hour. Power $\approx 32\%$. We catch the effect only 1 in 3 times—low power because 3 is only 1.5 standard errors from zero.
- **At $\beta^* = 6$:** Power $\approx 85\%$. The effect is now 3 standard errors from zero—large enough to detect reliably.
- **The curve is symmetric:** a negative effect of -3 is just as hard (or easy) to detect as $+3$.
- **The 80% dashed line** is a conventional benchmark for "adequate power" used in study design.

Why does the curve rise as $|\beta^*|$ grows? Because a larger true effect shifts the distribution of Z further from zero (larger $\delta = |\beta^*|/\sigma_{\hat{\beta}}$), placing more probability mass beyond the critical values ± 1.96. In the error-types diagram above, this is equivalent to moving the right-panel bell curve further to the right.


Power function: for each possible true effect, how often do we detect it?
sigma_hat = 2, alpha = 0.05. Red dots = the two values computed in Exercise 1.
100%
80%
60%
40%
20%
Power = P(reject H0)
beta*=6 => Pov
80% benchmark
beta*=3 => Power = 32%
beta*=0:
Power=alpha
floor = alpha = 5%
−5
0
5
True wage effect beta* ($/hour)

Part 2: Exercises *(25 min)*

Exercise: A Complete Example (25 min)

The Setting.

A researcher wants to know whether a job-training programme raises hourly wages. She estimates the effect with OLS and obtains $\hat{\beta}_1$ (dollars/hour). Under SLR.1–SLR.6 the estimator is normally distributed:

$$\hat{\beta}_1 \sim N\big(\beta_1,\ \sigma^2_{\hat{\beta}_1}\big) \qquad \text{with known } \sigma_{\hat{\beta}_1} = 2 \text{ \$/hour.}$$

She tests $H_0\colon \beta_1 = 0$ (no effect) against $H_1\colon \beta_1 \neq 0$ at $\alpha = 0.05$.

The test statistic is $Z = \hat{\beta}_1 / 2$, which follows $N(0,1)$ under H_0.

Part A — Theoretical Calculations

(a) Decision rule and critical values. Derive the values of $\hat{\beta}_1$ that lead to rejection of H_0.

Solution

Under H_0, the test statistic $Z = \hat{\beta}_1/\sigma_{\hat{\beta}_1}$ follows a standard normal distribution:

$$Z = \frac{\hat{\beta}_1}{2} \sim N(0,1) \quad \text{when } \beta_1 = 0.$$

We reject H_0 at $\alpha = 0.05$ whenever Z falls in the rejection region:

$$|Z| > z_{\alpha/2} = z_{0.025} = 1.96$$

Translating back to $\hat{\beta}_1$:

$$\left|\frac{\hat{\beta}_1}{2}\right| > 1.96 \iff |\hat{\beta}_1| > 1.96 \times 2 = 3.92 \text{ \$/hour}$$

Decision rule: reject H_0 if $\hat{\beta}_1 > 3.92$ or $\hat{\beta}_1 < -3.92$.

(b) Size and Type I error. What is the size of this test? Compute it explicitly and interpret in context.

Solution

The size is the probability of Type I error: rejecting H_0 when it is true ($\beta_1 = 0$). Since $Z \sim N(0,1)$ under H_0:

$$\text{Size} = P(|Z| > 1.96 \mid \beta_1 = 0) = P(Z > 1.96) + P(Z < -1.96) = 0.025 + 0.025 = \mathbf{0.05}$$

In context: if the training programme truly has no effect on wages, and we ran this study 100 times with different random samples, we would wrongly conclude that it works in approximately 5 of those studies—purely due to random variation in the data. We tolerate this 5% false-positive rate by choice

($\alpha = 0.05$).

(c) Power and Type II error. Suppose the programme *actually* raises wages by $\beta_1^* = 3$ \$/hour.

(i) Compute the **power** step by step.

(ii) Compute the **Type II error probability**.

(iii) Interpret both numbers in context.

Solution

(i) Power — step by step.
Step 1. When the truth is $\beta_1^* = 3$, the OLS estimator has distribution:

$$\hat{\beta}_1 \sim N(3,\ 4) \quad \Rightarrow \quad Z = \frac{\hat{\beta}_1}{2} \sim N\left(\frac{3}{2},\ 1\right) = N(1.5,\ 1)$$

The test statistic is no longer centred at zero—it is shifted right by $\delta = \beta_1^*/\sigma_{\hat{\beta}_1} = 3/2 = 1.5$.
Step 2. Power is the probability of falling in the rejection region under this shifted distribution:

$$\text{Power}(\beta_1^* = 3) = P(Z > 1.96 \mid Z \sim N(1.5, 1)) + P(Z < -1.96 \mid Z \sim N(1.5, 1))$$

Step 3. Standardise each term. If $Z \sim N(1.5, 1)$, then $W = Z - 1.5 \sim N(0, 1)$, so:

$$P(Z > 1.96) = P(W > 1.96 - 1.5) = P(W > 0.46) = 1 - \Phi(0.46) = 1 - 0.677 = 0.323$$
$$P(Z < -1.96) = P(W < -1.96 - 1.5) = P(W < -3.46) = \Phi(-3.46) \approx 0.0003$$

Step 4. Combine:

$$\text{Power}(3) = 0.323 + 0.0003 \approx \mathbf{0.323} = \mathbf{32.3}\%$$

(ii) Type II error. The Type II error is the complement of power (in the column "H_0 false"):

$$P(\text{Type II}) = 1 - \text{Power}(3) = 1 - 0.323 = \mathbf{0.677} = \mathbf{67.7}\%$$

(iii) Interpretation. The programme raises wages by \$3/hour, but our test detects this real effect only 32% of the time. In 68% of studies we would fail to reject H_0 and wrongly conclude the programme is ineffective—a Type II error. The power is low because \$3/hour is only 1.5 standard errors away from zero; the two hypotheses are hard to tell apart with this level of precision.

(d) Summary table. Fill in all four cells with the **probabilities** for this specific example ($\sigma_{\hat{\beta}_1} = 2$, $\alpha = 0.05$, $\beta_1^* = 3$).

	H_0 **true** ($\beta_1 = 0$)	H_0 **false** ($\beta_1^* = 3$)
Reject H_0		
Fail to reject H_0		

Solution

	H_0 **true** $(\beta_1 = 0)$	H_0 **false** $(\beta_1^* = 3)$
Reject H_0	**Type I error** $= \alpha = 0.05$	**Power** $= 0.323$
Fail to reject H_0	Correct $= 1 - \alpha = 0.95$	**Type II error** $= 1 - 0.323 = 0.677$

Key relationship: each column sums to 1 (the two outcomes are exhaustive). Type I error and Power live in *different columns*—they describe different states of the world. Reducing α (tighter Type I control) pushes the critical values outward, which makes it harder to reject H_0 and therefore *increases* Type II error. The only way to reduce both errors simultaneously is to increase the precision of $\hat{\beta}_1$ (i.e., larger sample size n).

(e) What if the true effect were larger? Compute power when $\beta_1^* = 6$ \$/hour, showing all steps. Explain why power is higher than in (c).

Solution

Step 1. New signal-to-noise ratio: $\delta = 6/2 = 3$. The distribution of Z is now $N(3, 1)$.
Step 2–3. Standardise:

$$P(Z > 1.96 \mid Z \sim N(3,1)) = P(W > 1.96 - 3) = P(W > -1.04) = \Phi(1.04) = 0.851$$
$$P(Z < -1.96 \mid Z \sim N(3,1)) = P(W < -1.96 - 3) = \Phi(-4.96) \approx 0$$

Step 4.

$$\text{Power}(6) = 0.851 + 0 \approx \mathbf{0.851} = \mathbf{85.1\%}$$

Type II error $= 1 - 0.851 = 14.9\%$.
Why is power higher? When $\beta_1^* = 6$, the distribution of Z is centred at $\delta = 3$—three standard errors from zero. Most of its mass now lies to the right of the critical value 1.96, making rejection the most likely outcome. Compared to $\beta_1^* = 3$ (where $\delta = 1.5$), the distribution has moved further from the threshold, and detection is much easier.

Part B — Simulation

Goal. We verify the numbers from Part A by simulation using the same job-training context.

The idea: imagine running the wage study 10,000 times—drawing a new random sample of workers each time and computing $\hat{\beta}_1$. We do this twice: once in a world where the programme has no effect (**Scenario A**: $\beta_1 = 0$) and once where it truly raises wages by \$3/hour (**Scenario B**: $\beta_1^* = 3$).

In each simulated study we apply the same decision rule: reject H_0 if $|\hat{\beta}_1| > 3.92$.

- In Scenario A, the fraction of rejections estimates the **size** (should be ≈ 0.05).
- In Scenario B, the fraction of rejections estimates the **power** (should be ≈ 0.323).
- In Scenario B, the fraction of *non*-rejections estimates the **Type II error** (should be ≈ 0.677).

```r
set.seed(42)
sigma_hat <- 2             # SE of beta_hat (same as Part A)
z_crit    <- qnorm(0.975) # 1.96  =>  reject when |beta_hat| > 1.96 * 2 = 3.92
n_sim     <- 10000        # number of simulated studies

# ---------------------------------------------------------------
# Scenario A: H0 is TRUE -- programme has NO effect (beta_1 = 0)
# Each draw is one hypothetical study: a new sample of workers,
# a new OLS estimate.  Here beta_hat ~ N(0, sigma_hat^2).
# ---------------------------------------------------------------
bhat_A   <- rnorm(n_sim, mean = 0, sd = sigma_hat)
reject_A <- abs(bhat_A / sigma_hat) > z_crit   # TRUE = wrongly rejected H0

cat("--- Scenario A: programme has NO effect (beta_1 = 0) ---\n")
```

```
## --- Scenario A: programme has NO effect (beta_1 = 0) ---
```

```r
cat("  Reject H0 (Type I error):  ", round(mean(reject_A),  3),
    "  [theory: 0.050]\n")
```

```
##   Reject H0 (Type I error):   0.051   [theory: 0.050]
```

```r
cat("  Do not reject (correct):   ", round(mean(!reject_A), 3),
    "  [theory: 0.950]\n\n")
```

```
##   Do not reject (correct):    0.949   [theory: 0.950]
```

```r
# ---------------------------------------------------------------
# Scenario B: H0 is FALSE -- true effect is beta_1* = 3 $/hour
# Now beta_hat ~ N(3, sigma_hat^2).  The distribution has shifted.
# ---------------------------------------------------------------
bhat_B   <- rnorm(n_sim, mean = 3, sd = sigma_hat)
reject_B <- abs(bhat_B / sigma_hat) > z_crit   # TRUE = correctly rejected H0

cat("--- Scenario B: programme raises wages by $3/hour (beta_1* = 3) ---\n")
```

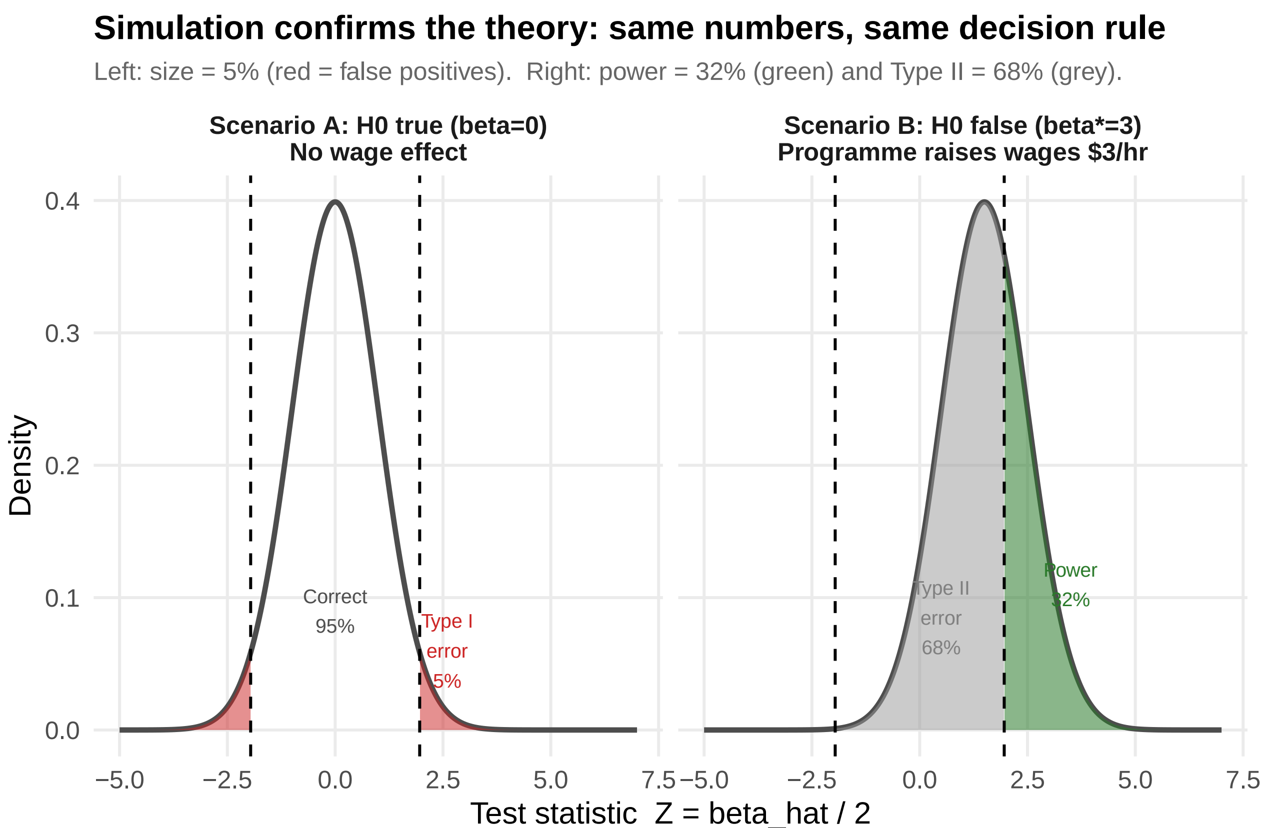
```
## --- Scenario B: programme raises wages by $3/hour (beta_1* = 3) ---
```

```r
cat("  Reject H0 (Power):         ", round(mean(reject_B),  3),
    "  [theory: 0.323]\n")
```

```
##   Reject H0 (Power):          0.326   [theory: 0.323]
```

```r
cat("  Do not reject (Type II):   ", round(mean(!reject_B), 3),
    "  [theory: 0.677]\n")
```

```
##   Do not reject (Type II):    0.674   [theory: 0.677]
```



Solution

Reading the results:

- **Scenario A — size** ≈ 0.05**:** in 10,000 studies where the programme has no effect, we wrongly rejected H_0 in about 5% of them. This is exactly the Type I error rate we chose ($\alpha = 0.05$). The result confirms that our critical value of ± 1.96 is correctly calibrated.
- **Scenario B — power** ≈ 0.323**:** in 10,000 studies where the programme genuinely raises wages by $3/hour, we correctly detected the effect in only about 32% of them—matching the analytical calculation in Part A(c). This is low power: a real, meaningful wage gain goes undetected most of the time.
- **Scenario B — Type II error** ≈ 0.677**:** in the remaining 68% of Scenario B studies, we failed to reject H_0 even though the programme works. These are false negatives—studies that would lead a policy-maker to cancel a beneficial programme.
- **Connection to the figure:** the two panels of the plot correspond exactly to the two scenarios. The left panel (Scenario A) shows the $N(0, 1)$ distribution of Z; the red tails are the 5% false-positive rate. The right panel (Scenario B) shows the shifted $N(1.5, 1)$ distribution; the green area is power and the grey area is the Type II error.